### 2.2.5 ▶ Efficiency of a Carnot Engine

The Carnot Engine is a perfectly reversible engine operating between a hot and a cold temperature. As shown in 2.2.4, the efficiency of any such engine is the same for two reservoirs of given temperatures and is the upper bound for efficiency of an engine operating between the temperatures. A common way to construct the Carnot Engine is as follows (and as depicted in the diagram):

1. An isothermal expansion at hot temperature $T_H$
2. An adiabatic expansion
3. An isothermal compression at cold temperature $T_C$
4. An adiabatic compression

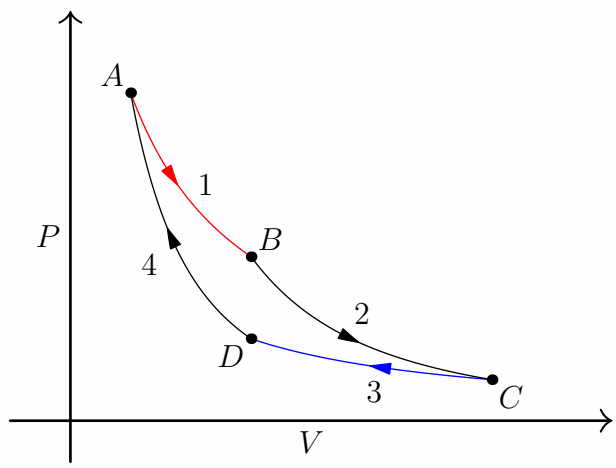


**Fig. 2.5:** Carnot Cycle, with two adiabatic and two isothermal processes. The hot temperature $T_H$ is depicted in red while the cold temperature $T_C$ is depicted in blue.

To find the efficiency of the engine, consider first the definition of engine efficiency, which is the ratio of the work output and the heat input. We can thus break each process down and see where heat is being inputted and where work is extracted (note that all of the following values are magnitudes).

1. Heat $Q_H$ is inputted and work $W_1$ is extracted. Since $\Delta U = 0$, $Q_H = W_1$.[8]
2. No heat is exchanged but work $W_2$ is extracted.[9]
3. Heat $Q_C$ is extracted and work $W_3$ is inputted. Since $\Delta U = 0$, $Q_C = W_3$.
4. No heat is exchanged but work $W_4$ is inputted.

[8]Note here that the volume increases at a positive pressure, meaning that the quantity $W = -P\Delta V$ is negative, but the work that the gas does is positive. Why? It's because the quantity $-P\Delta V$ is the work done *on* the gas, not *by* the gas, i.e. it gives the work inputted into the gas, not the work extracted. An intuitive way of thinking of it is like this: it takes effort to compress a piston and decrease its volume (positive work input) and if released from a compressed state, the piston springs outwards (positive work output).

[9]Again, $W_2$ is a positive value despite there being an increase in volume.

By conservation of energy, the work output of the engine is $Q_H - Q_C$, meaning that the efficiency of the engine is

$$\epsilon_{carnot} = \frac{W_{out}}{Q_H} = \frac{Q_H - Q_C}{Q_H}. \tag{2.32}$$

Since $Q_H = W_1$, we can find $Q_H$ by solving for the area under process 1:

$$\begin{aligned} Q_H = W_1 &= \int_{V_A}^{V_B} PdV \\ &= nRT_H \int_{V_A}^{V_B} \frac{dV}{V} \\ &= nRT_H \ln\left(\frac{V_B}{V_A}\right) \end{aligned} \tag{2.33}$$

where $V_A$ and $V_B$ are the volumes of the gas at state $A$ and $B$, respectively. By the same logic,

$$Q_C = W_3 = \int_{V_D}^{V_C} PdV = nRT_C \ln\left(\frac{V_C}{V_D}\right) \tag{2.34}$$

Plugging eq. 2.33 and 2.34 into 2.32, we have

$$\epsilon_{carnot} = 1 - \frac{T_C}{T_H} \cdot \frac{\ln(V_B/V_A)}{\ln(V_C/V_D)} \tag{2.35}$$

From the derivation in 2.2.3 (specifically eq. 2.29), we can relate the volumes between the start and end points of the adiabatic processes

$$\begin{aligned} \frac{T_H}{T_C} = \left(\frac{V_C}{V_B}\right)^{\gamma-1} &= \left(\frac{V_D}{V_A}\right)^{\gamma-1} \\ \frac{V_C}{V_D} &= \frac{V_B}{V_A}. \end{aligned}$$

With this result, eq. 2.35 simply reduces to

$$\boxed{\epsilon_{carnot} = 1 - \frac{T_C}{T_H}}. \tag{2.36}$$

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